# 27.6: Prokaryotes have both beneficial and harmful impacts on humans

Although the best-known prokaryotes tend to be the bacteria that cause human illness, these pathogens represent only a small fraction of prokaryotic species. Many other prokaryotes have positive interactions with people, and some play essential roles in agriculture and industry.

## Mutualistic Bacteria

As is true for many other eukaryotes, human well-being can depend on mutualistic prokaryotes. For example, our intestines are home to an estimated 500–1,000 species of bacteria; collectively, their cells outnumber all human cells in the body by a factor of ten. Different species live in different portions of the intestines, and they vary in their ability to process different foods. Many of these species are mutualists, digesting food that our own intestines cannot break down. The genome of one of these gut mutualists, *Bacteroides thetaiotaomicron*, includes a large array of genes involved in synthesizing carbohydrates, vitamins, and other nutrients needed by humans. Signals from the bacterium activate human genes that build the network of intestinal blood vessels necessary to absorb nutrient molecules. Other signals induce human cells to produce antimicrobial compounds to which *B. thetaiotaomicron* is not susceptible. This action may reduce the population sizes of other, competing species, thus potentially benefiting both *B. thetaiotaomicron* and its human host.

## Pathogenic Bacteria

All the pathogenic prokaryotes known to date are bacteria, and they deserve their negative reputation. Bacteria cause about half of all human diseases. For example, more than 1.5 million people die each year of the lung disease tuberculosis, caused by *Mycobacterium tuberculosis*. And another 2 million people die each year from diarrheal diseases caused by various bacteria.

Some bacterial diseases are transmitted by other species, such as fleas or ticks. In the United States, the most widespread pest-carried disease is Lyme disease, which infects about 300,000 people each year according to a recent CDC estimate **(Figure 27.21)**. Caused by a bacterium carried by ticks, Lyme disease can result in debilitating arthritis, heart disease, nervous disorders, and death if untreated.

**Figure 27.21**

```
Lyme disease.
```

Photographs of a tick, the spirochete Borrelia burgdorferi, and the rash that can appear at the site of a tick's bite.

*Ticks in the genus Ixodes spread the disease by transmitting the spirochete Borrelia burgdorferi (colorized SEM). A rash may develop at the site of the tick's bite; the rash may be large and ring-shaped (as shown) or much less distinctive.*

Tick photographed on a human finger for scale reference; appears as roughly one seventh diameter of the finger. Spirochete Borrelia burgdorferi has a spiraling, string-like shape (SEM, scale 5 micrometers). Rash on a person's shoulder appears as a bulls-eye with a dark center and rings radiating outward.

Pathogenic prokaryotes usually cause illness by producing poisons, which are classified as exotoxins or endotoxins. **Exotoxins** are proteins secreted by certain bacteria and other organisms. Cholera, a dangerous diarrheal disease, is caused by an exotoxin secreted by the proteobacterium *Vibrio cholerae*. The exotoxin stimulates intestinal cells to release chloride ions into the gut, and water follows by osmosis. In another example, the potentially fatal disease botulism is caused by botulinum toxin, an exotoxin secreted by the gram-positive bacterium *Clostridium botulinum* as it ferments various foods, including improperly canned meat, seafood, and vegetables. Like other exotoxins, the botulinum toxin can produce disease even if the bacteria that manufacture it are no longer present when the food is eaten.

**Endotoxins** are lipopolysaccharide components of the outer membrane of gram-negative bacteria. In contrast to exotoxins, endotoxins are released only when the bacteria die and their cell walls break down. Endotoxin-producing bacteria include species in the genus *Salmonella*, such as *Salmonella typhi*, which causes typhoid fever. You might have heard of food poisoning caused by other *Salmonella* species that can be found in poultry and some fruits and vegetables.

Finally, horizontal gene transfer can spread genes associated with virulence, turning normally harmless bacteria into potent pathogens. *E. coli*, for instance, is ordinarily a harmless symbiont in the human intestines, but pathogenic strains that cause bloody diarrhea have emerged. One of the most dangerous strains, O157:H7, is a global threat; in the United States alone, there are around 75,000 cases of O157:H7 infection per year, often from contaminated beef or produce. Scientists have discovered over 1,000 genes in O157:H7 that have no counterpart in harmless *E. coli* strains. Some of the genes found only in O157:H7 are associated with virulence and were likely transferred by phage-mediated horizontal gene transfer (transduction) from pathogenic bacterial species.

## Antibiotic Resistance

Since their initial use in the 1940s, antibiotics have saved many lives and reduced the incidence of disease caused by pathogenic bacteria. Unfortunately, as **Figure 27.22** shows, resistance to antibiotics has evolved in many bacteria, often rapidly. Compounding this problem, the discovery of new antibiotics has not kept pace with the rate at which resistance has evolved in bacteria. We now face a major public health concern: For every antibiotic now in use, at least one species of bacteria has developed resistance to it.

**Figure 27.22**

```
The rise of antibiotic resistance.
```

A chart lists a variety of antibiotics, the years they were discovered, when medical usage began, and when resistance was observed. The range is from 1930 to present day. All antibiotics given have had resistance observed.

As illustrated by the examples shown here, bacteria have developed resistance to every antibiotic currently in clinical use, often within a few years. It took more than 50 years for resistance to develop to colistin, most likely because this antibiotic has toxic side effects in humans and hence was used rarely (as a treatment of last resort).

**VISUAL SKILLS***Based on this diagram, identify the antibiotic for which resistance appeared* before *the antibiotic began to be used medicinally.*

*For suggested answer, see*Appendix A.**

The rise of antibiotic resistance is driven by several factors. The rapid reproduction of bacteria enables cells carrying resistance genes to quickly produce large numbers of resistant offspring. Thus, when humans use antibiotics in medical or agricultural settings, natural selection can cause the fraction of a bacterial population carrying genes for antibiotic resistance to increase rapidly. Resistance genes also can spread within and among bacterial species by horizontal gene transfer. As a result, bacterial strains that are resistant to multiple antibiotics are becoming more common, making the treatment of certain bacterial infections very difficult.

Consider tuberculosis, currently the second leading cause of death by infectious disease worldwide (after COVID-19). The recent rise of drug-resistant strains of the tuberculosis bacterium (*Mycobacterium tuberculosis*) threatens decades of progress in combating tuberculosis. “Superbug” tuberculosis strains came to public notice in 2006, when an outbreak in a rural area of South Africa killed 52 of the 53 patients infected with what is now called extensively drug-resistant tuberculosis (XDR-TB). As of 2017, more than 100 countries had confirmed cases of XDR-TB. There are few treatments for XDR-TB, and those we have are difficult to administer and more toxic than are treatments for standard strains of the tuberculosis bacterium.

*M. tuberculosis* is not the only bacterium in which resistance to multiple antibiotics has evolved. Highly resistant strains have also been found in species that cause potentially lethal infections of the skin, lungs, digestive tract, and other human organs. Even so, recent discoveries give cause for hope. In one such study, researchers used a metagenomic approach to sequence genes from soil bacteria that could not be grown in the laboratory. They found that some of these genes encoded a new class of antibiotics, which have been named *malacidins*. Malacidins have been effective against multidrug-resistant gram-positive pathogens, including clone USA300 **(Figure 27.23)**, a highly virulent strain of methicillin-resistant *Staphylococcus aureus* (MRSA; see Figure 22.14). Researchers also have developed virus-like particles capable of targeting (and killing) particular species of multidrug-resistant bacteria. Novel approaches for growing soil bacteria in the laboratory also have led to the discovery of new antibiotics, as you can explore in the **Scientific Skills Exercise**.

**Figure 27.23**

```
Effectiveness of malacidin against MRSA clone USA300.
```

Two graphs plot MRSA abundance on a log scale versus time in hours.

The control (no antibiotics) rises from (0, 2.5) to (24, 5.5), then rises to (72, 7). The treatment with malacidin falls from (0, 2.5) to (24, 0), then goes to (72, 0). All values are estimated.

Treatment with malacidin rapidly cleared a skin infection in rats caused by clone USA300, a highly virulent strain of methicillin-resistant Staphylococcus aureus (MRSA). Malacidin was as effective as daptomycin, an antibiotic used to treat life-threatening infections.

## Scientific Skills Exercise

## *Calculating and Interpreting Means and Standard Errors*

### AP Science Practices 3, 4, 6

**Can Antibiotics Obtained from Soil Bacteria Help Fight Drug-Resistant Bacteria?** Soil bacteria synthesize antibiotics, which they use against species that attack or compete with them. To date, these species have been inaccessible as sources for new medicines because 99% of soil bacteria cannot be grown using standard laboratory techniques. To address this problem, researchers developed a method in which soil bacteria grow in a simulated version of their natural environment; this led to the discovery of a new

antibiotic, teixobactin. In this exercise, you'll calculate means and standard errors from an experiment that tested teixobactin's effectiveness against MRSA (methicillin-resistant *Staphylococcus aureus*; see Figure 22.14).

Plastic chip used to grow soil bacteria.

***Plastic chip used to grow soil bacteria***

**How the Experiment Was Done** Researchers drilled tiny holes into a small plastic chip and filled the holes with a dilute aqueous solution containing soil bacteria and agar. The dilution had been calibrated so that only one bacterium was likely to grow in each hole. After the agar solidified, the chip was placed in a container containing the original soil; nutrients and other essential materials from the soil diffused into the agar, allowing the bacteria to grow.

After isolating teixobactin from a soil bacterium, researchers performed the following experiment: Mice infected with MRSA were given low (1 mg/kg) or high (5 mg/kg) doses of teixobactin or vancomycin, an existing antibiotic; in the control, mice infected with MRSA were not given an antibiotic. After 26 hours, researchers sampled infected mice and estimated the number of *S. aureus* colonies in each sample. Results were reported on a log scale; note that a decrease of 1.0 on this scale reflects a tenfold decrease in MRSA abundance.

**Data from the Experiment**

A table lists treatment, dose, log of the number of colonies, and mean.

**Data from** L. Ling et al. A new antibiotic kills pathogens without detectable resistance, *Nature* 517:455–459 (2015).

The table is as follows:

| Treatment | Dose in milligram per kilogram | Log of number of colonies | Mean |
|---|---|---|---|
| Control | not applicable | 9.0, 9.5, 9.0, 8.9 | blank |
| Vancomycin | 1 | 8.5, 8.4, 8.2 | blank |
| Vancomycin | 5 | 5.3, 5.9, 4.7 | blank |
| Teixobactin | 1 | 8.5, 6.0, 8.4, 6.0 | blank |
| Teixobactin | 5 | 3.8, 4.9, 5.2, 4.9 | blank |

**INTERPRET THE DATA**

- The mean ($\bar{x}$) of a variable is the sum of the data values divided by the number of observations (n):

$\bar{x}=\sum xin$

- In this formula, xi is the value of the ith observation of the variable (observation number "i" out of a total of n such observations); the Σ symbol indicates that the n values of x are to be added together. Calculate the mean for each treatment.
- Use your results from question 1 to evaluate the effectiveness of vancomycin and teixobactin.
- The variation found in a set of data can be estimated by the standard deviation, s:

$s=\sum(xi-\bar{x})2n-1$

- Calculate the standard deviation for each treatment.
- The standard error (SE), which indicates how greatly the mean would likely vary if the experiment was repeated, is calculated as:

$SE=sn$

- As a rough rule of thumb, if an experiment were to be repeated, the new mean typically would lie within two standard errors of the original mean (that is, within the range $\bar{x}\pm 2SE$. Calculate $\bar{x}\pm 2SE$ for each treatment, determine whether these ranges overlap, and interpret your results.

**Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

## Prokaryotes in Research and Technology

On a positive note, we reap many benefits from the metabolic capabilities of both bacteria and archaea. For example, people have long used bacteria to convert milk to cheese and yogurt. Bacteria are also used in the production of beer and wine, pepperoni, fermented cabbage (sauerkraut), and soy sauce. In recent decades, our greater understanding of prokaryotes has led to an explosion of new applications in biotechnology. Examples include the use of *E. coli* in gene cloning (see Figure 20.4) and the use of DNA polymerase from *Pyrococcus furiosus* in the PCR technique (see Figure 20.8). Through genetic engineering, we can modify bacteria to produce vitamins, antibiotics, hormones, and other products (see Concept 20.1).

Recently, the prokaryotic CRISPR-Cas system, which helps bacteria and archaea defend against attack by viruses (see Figure 19.7), has been developed into a powerful new tool for altering genes in virtually any organism. The genomes of many prokaryotes contain short DNA repeats, called CRISPRs, that interact with

proteins known as the Cas (CRISPR-associated) proteins. Cas proteins, acting together with “guide RNA” made from the CRISPR region, can cut any DNA sequence to which they are directed. Scientists have been able to exploit this system by introducing a Cas protein (Cas9) to guide RNA into cells whose DNA they want to alter (see Figure 17.28). Among other applications, this CRISPR-Cas9**system** has already opened new lines of research on HIV, the virus that causes AIDS **(Figure 27.24)**. While the CRISPR-Cas9 system can potentially be used in many different ways, care must be taken to guard against the unintended consequences that could arise when applying such a new and powerful technology.

**Figure 27.24**

```
    CRISPR: opening new avenues of research for treating HIV
infection.
```

Part a. Controlled cells. Their green color indicates infection by H I V. Part b. Experimental cells. These cells were treated with a CRISPR-Cas9 system that targets H I V. There is little to no green visible.

**(a)** In laboratory experiments, untreated (control) human cells were susceptible to infection by HIV, the virus that causes AIDS. **(b)** In contrast, cells treated with a CRISPR-Cas9 system that targets HIV were resistant to viral infection. The CRISPR-Cas9 system was also able to remove HIV proviruses (see Figure 19.8) that had become incorporated into the DNA of human cells.

Another valuable application of bacteria is to reduce our use of petroleum. Consider the plastics industry. Globally, each year more than 1 trillion (1,000,000,000,000) pounds of plastic are produced from petroleum and used to make toys, storage containers, soft drink bottles, and many other items. Many of these products end up as plastic waste that contaminates natural habitats and degrades slowly, creating environmental problems (see Concept 56.4). One promising avenue for addressing these problems is to use bacteria that produce natural plastics **(Figure 27.25)**. For example, some bacteria synthesize a type of polymer known as PHA (polyhydroxyalkanoate), which they use to store chemical energy. The PHA can be extracted, formed into pellets, and used to make durable, yet biodegradable, plastics. Researchers are also seeking to reduce the use of petroleum and other fossil fuels by engineering bacteria that can produce ethanol from various forms of biomass, including agricultural waste, switchgrass, and corn.

**Figure 27.25**

```
    Bacteria synthesizing and storing PHA, a component of
```

```
biodegradable plastics.
```

Round and oval-shaped bacteria have visible pockets or spaces inside them.

Another way to harness prokaryotes is **bioremediation**, the use of organisms to remove pollutants from soil, air, or water. For example, anaerobic bacteria and archaea decompose the organic matter in sewage, converting it to material that can be used as landfill or fertilizer after chemical sterilization. Other bioremediation applications include cleaning up oil spills **(Figure 27.26)** and precipitating radioactive material (such as uranium) out of groundwater.

**Figure 27.26**

```
    Bioremediation of an oil spill.
```

Men spray fertilizer on an oil spill along a shoreline.

*Spraying fertilizer stimulates the growth of native bacteria that metabolize oil, increasing the speed of the breakdown process up to fivefold.*

The usefulness of prokaryotes largely derives from their diverse forms of nutrition and metabolism. All this metabolic versatility evolved prior to the appearance of the structural novelties that heralded the evolution of eukaryotic organisms, discussed in the rest of this unit.

---